

Smog City: Visualizing the Invisible Threat

Air Quality Index (AQI) Analysis of Indian Cities — Course: Data Science and Visualization (24ADI204) — Domain: Environment, Climate & Energy



Problem Statement

Air pollution is one of India's largest environmental and public-health challenges. Persistent poor air quality harms respiratory and cardiovascular health, reduces visibility, and diminishes life quality.

Our project analyses AQI from major Indian cities to reveal trends, pinpoint key pollutant drivers, and surface evidence that can guide policy and public action.



Project Objectives

Data Analysis

Perform city-level and temporal AQI analysis to surface trends and extremes.

Data Quality

Handle missing values and outliers with principled imputation and detection strategies.

Feature Insights

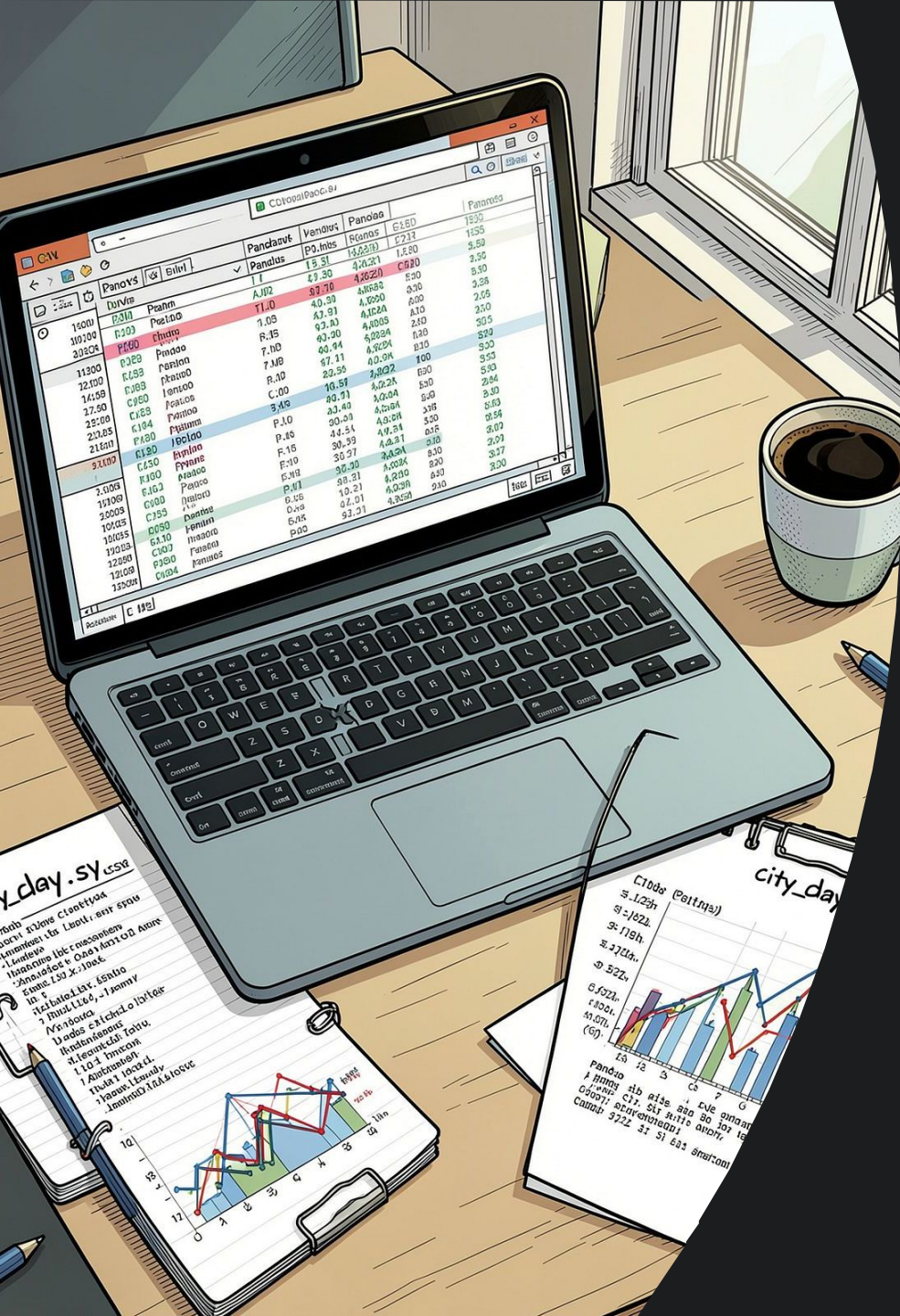
Identify pollutants (PM2.5, PM10, NO₂, SO₂, CO, O₃, VOCs) that drive AQI variability.

Dimensionality Reduction

Use PCA to summarise correlated pollutant signals into principal components for interpretation.

Storytelling

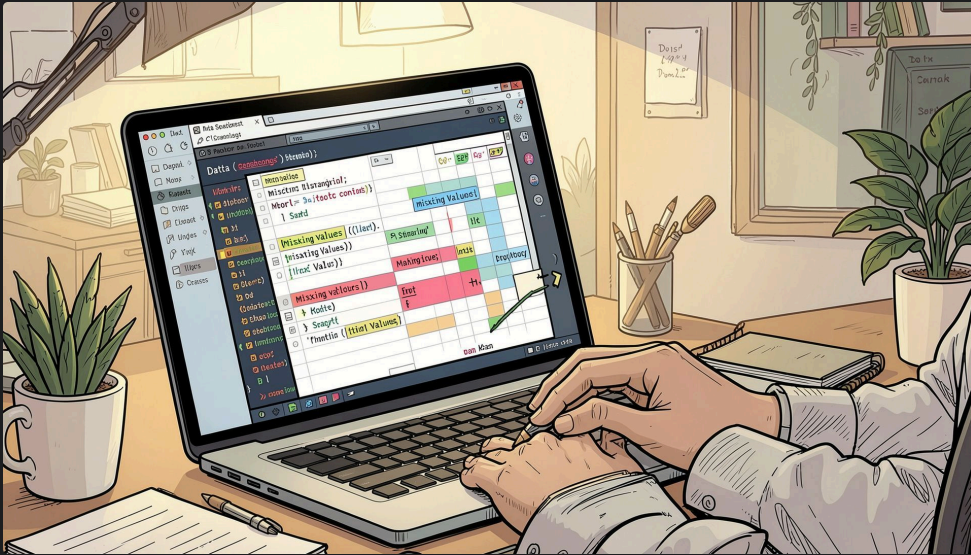
Deliver an interactive dashboard that communicates findings and supports exploration.



Dataset — Overview

Source: Kaggle — "Air Quality Data in India (2015–2020)" — Primary file: `city_day.csv`. ([link for reference](#))

Key features: City, Date, AQI, PM2.5, PM10, NO₂, SO₂, CO, O₃, Benzene, Toluene, AQI Category. Data spans multi-year daily observations across major Indian cities.



Methodology — Steps

1. Data collection & validation
2. Data cleaning: parse dates, normalise city names
3. Missing value handling: imputation strategies per pollutant
4. Outlier detection: statistical and domain-based thresholds
5. Exploratory Data Analysis: trends, seasonal patterns
6. Feature scaling and PCA
7. Dashboard development with Plotly + Streamlit
8. Story-driven visualisations and policy recommendations

Tools & Technologies

Primary stack: Python, Pandas, NumPy, Scikit-learn for modelling and PCA; Matplotlib & Seaborn for static plots; Plotly and Streamlit for interactive visualisations and dashboarding; GitHub for version control and reproducibility.

- Use reproducible notebooks and data provenance to ensure transparency for research and coursework.

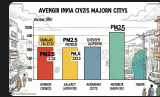


Expected Visualizations



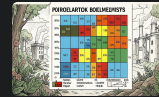
AQI Trend Analysis

Multi-year line charts with seasonal decomposition to reveal annual cycles and extreme episodes.



Pollution by City

City-wise comparisons highlighting hotspots and relative pollutant burdens.



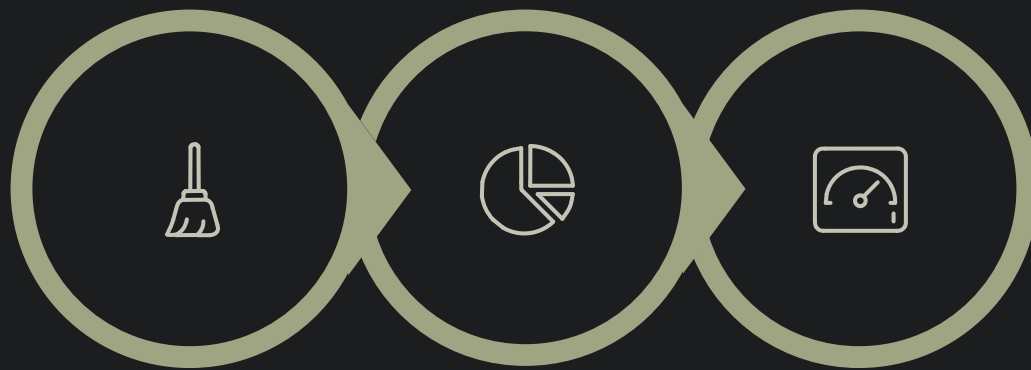
Correlation Heatmap

Shows co-variation among pollutants to inform PCA and source interpretation.



Interactive India Map

Choropleth/marker map for geographical exploration and time-slider playback.



**Clean &
Impute**

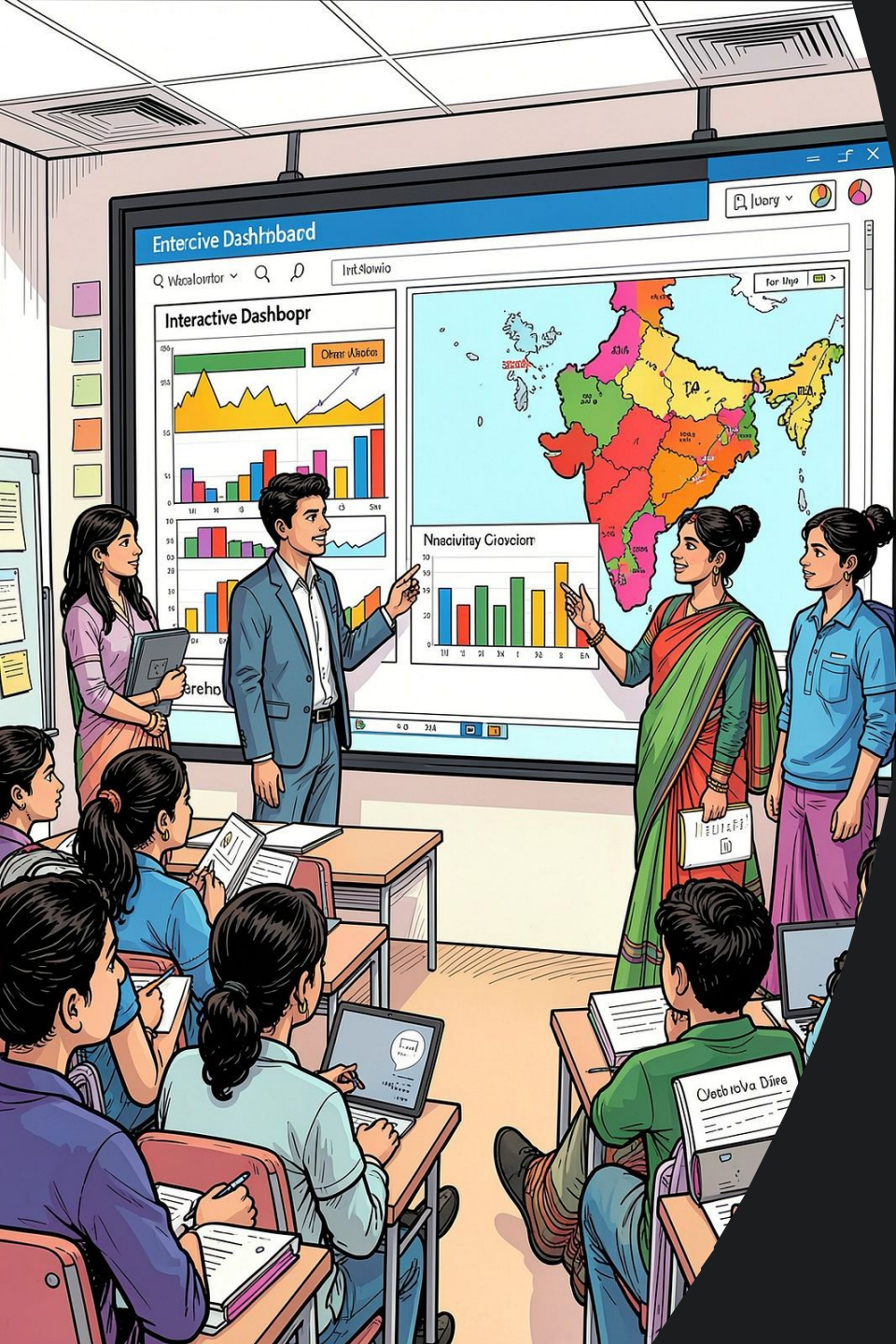
**Analyse &
Reduce**

**Build
Dashboard**

This workflow ensures data quality, extracts core pollutant signals via PCA, and packages insights into an interactive experience for researchers and decision-makers.

PCA & Interpretation

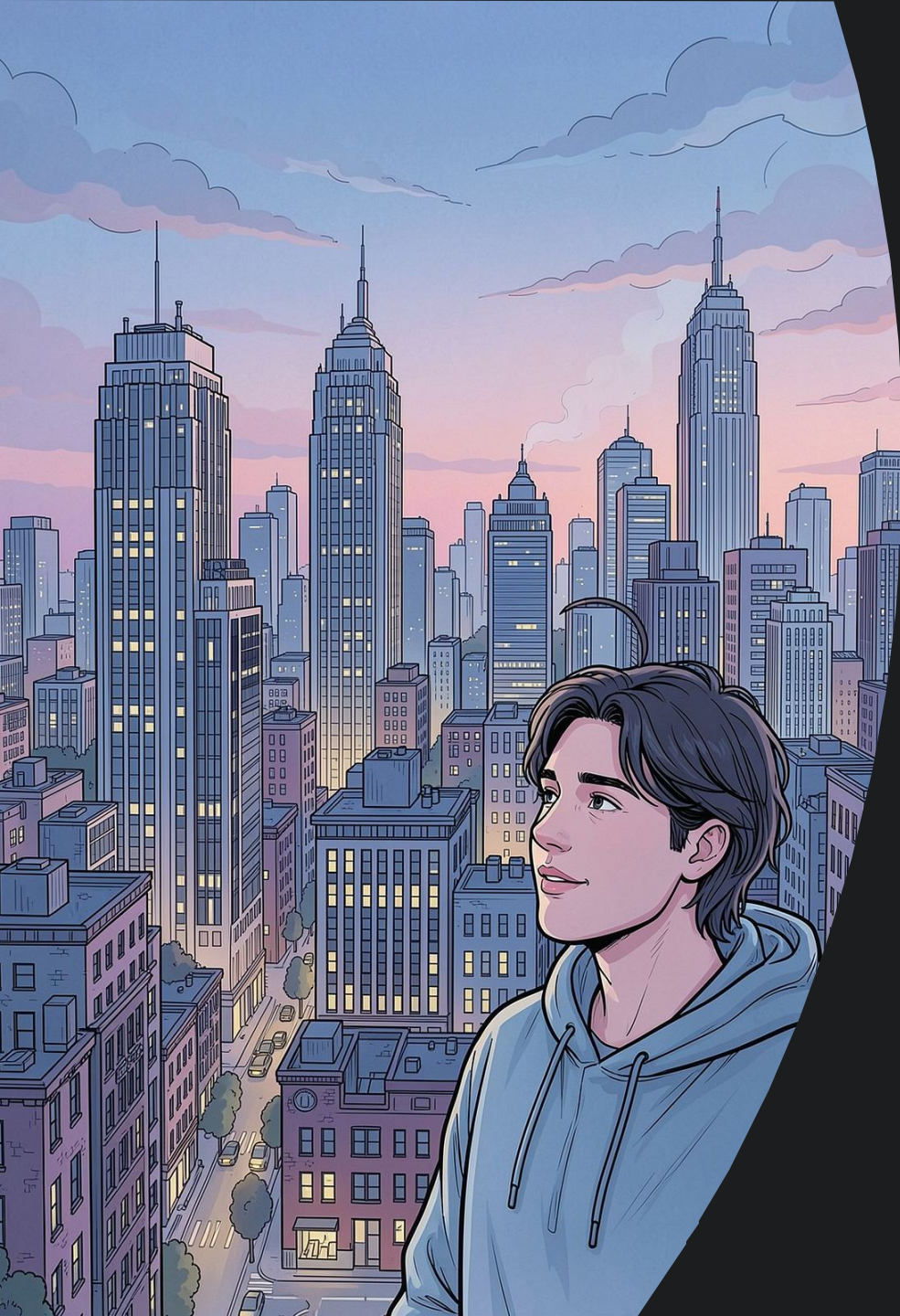
PCA will condense correlated pollutant variables into principal components representing shared sources (e.g., combustion vs. dust). We'll examine loadings to attribute components to likely emission sources and use PCA scores to cluster cities by pollutant signature.



Expected Outcomes & Impact

- Identify the most polluted cities and seasonal high-risk periods.
- Quantify which pollutants (PM2.5, PM10, NO₂, etc.) contribute most to AQI variance.
- Reveal pollutant source patterns via PCA and correlation analysis.
- Deliver an interactive dashboard for stakeholders to explore scenarios and trends.
- Provide concise, data-driven recommendations to inform local mitigation strategies.

i Outcome emphasises reproducible analysis, clear visual communication, and actionable recommendations.



Conclusion & Next Steps

This project transforms raw AQI observations into actionable insight: from cleaning and PCA to interactive dashboards that inform policy and public awareness. Next steps: refine imputation, validate PCA components with source data, and deploy the Streamlit dashboard with time-slider and city filters.

References: Kaggle dataset — Air Quality Data in India (2015–2020).
Course: 24ADI204.